A Model-Conditioned Preflight Protocol for Prior Selection in Node-Wise Scientific Latent Dynamics

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Abstract

Graph-prior gains in scientific latent dynamics are often reported as prediction improvements, but such gains do not by themselves show that the model used the candidate graph topology. We propose a model-conditioned preflight protocol that decomposes graph-prior efficacy into three operational attribution classes: (i) topology-aligned support, requiring agreement between rollout point-estimate ordering and audit-mode latent-metric separation against the spectrum-matched permuted graph (SMPG) control; (ii) generic regularization, where graph priors improve over no-prior training but the gains do not survive SMPG controls; and (iii) no effect. Applied across seven regimes including HO lattices, N-body distance graphs at three connectivities, real-world traffic (METR-LA), graph-heat and graph-wave lattices, spring-mass systems, and five rMD17 molecules—totaling 13 measurement cells—the protocol identifies one controlled audit-positive (topology-aligned support) case (the HO audit), eight generic-regularization cells spanning both N-body synthetic data and real rMD17 molecular-dynamics data, and four no-effect cells. Across the eight generic-regularization cells, effect sizes against the no-prior baseline range from 2.4 to 8.4, while effect sizes against the SMPG control range from (-2.9) to 0.9 with confidence-interval overlap in every cell. Thus, graph-vs-none improvements in our evidence would have been over-attributed without SMPG controls. The protocol is offered as a practical preflight workflow for prior selection and interpretation under a fixed model condition; it is not a new graph prior, a universal physics predictor, a causal discovery method, or a state-of-the-art benchmark.

1. Introduction

Scientific dynamics models increasingly combine learned latent-state evolution with auxiliary priors motivated by domain structure. In molecular dynamics, coupled oscillator systems, particle simulations, traffic networks, and graph-discretized physical fields, observations can often be written as trajectories X_t over N interacting nodes. Before fitting a full model, a practitioner may already have a plausible candidate graph, a temporal smoothness assumption, or another regularizer that appears scientifically reasonable. Such priors are attractive because they may stabilize long-horizon rollout prediction, improve sample efficiency, and encode structure that is difficult to infer from limited data alone. Their practical value, however, is not intrinsic to the prior alone. It depends on how the prior interacts

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with the encoder, transition model, latent dimension, prior placement, optimization setting, training budget, and rollout horizon.

In the settings tested here, this dependence makes graph-prior effects difficult to interpret. A Laplacian penalty on learned node-wise latents may improve rollout because the candidate graph captures meaningful physical or scientific structure. The same gain may also arise because the penalty imposes generic graph-frequency smoothing, because a spectrum-matched permuted graph control would have produced a similar effect, because calibrated temporal smoothing would have been sufficient, or because the apparent advantage is confined to a short budget before an unconstrained model catches up. A graph-aware encoder or transition model can further blur the interpretation by already encoding part of the relational bias. Consequently, in our evidence and protocol, a graph prior outperforming the no-prior baseline establishes practical utility but does not by itself establish topology-specific attribution.

From a computational-science perspective, this ambiguity is costly. In our workflow, sweeps over priors, graph constructions, regularization strengths, seeds, and budgets can still leave attribution unresolved if the comparison set is incomplete. What is often needed first is a smaller decision stage: should the candidate graph be carried into a larger run, should a cheaper graph-free alternative be preferred, or should stronger controls be introduced before any topology-specific attribution claim is made? We therefore frame prior choice as a preflight problem. Before committing to full-scale training, a structured low-cost screen should determine whether a prior is useful, what explains the gain, and what evidence would still be required for stronger interpretation.

This article develops that idea as a model-conditioned preflight protocol for prior selection in node-wise scientific latent dynamics. The protocol compares no-prior training, a candidate graph prior, a spectrum-matched permuted graph control, optional random-graph controls, and calibrated temporal smoothing. Audit mode, used when checkpointed latent traces are available, additionally examines whether predictive gains coincide with latent behavior aligned with the candidate graph. The contribution is methodological rather than architectural: the paper introduces a control-oriented workflow for prior selection and topology-specific attribution, and it uses a conservative label set to convert ambiguous prior effects into an actionable prior recommendation under a specified model condition. The manuscript does not introduce a new graph prior, claim universal physics prediction, frame the task as causal discovery, or present a state-of-the-art benchmark.

2. Related Work

This manuscript sits at the intersection of several active lines of research while remaining narrower in scope than any one of them. A first strand comes from geometric deep learning, where graph neural networks, graph-spectral constructions, and graph-based regularizers are used to encode relational inductive bias on non-Euclidean domains [1, 2]. Prior literature establishes graph structure as useful for prediction and representation learning; our narrower question is attribution under a fixed tested protocol. This paper instead asks: under a fixed latent dynamics model, when does a candidate graph prior merit further computation, and what controls are required before topology-specific attribution is justified?

A second strand concerns scientific and physical latent dynamics models, including world-model formulations that learn compact state representations and transition operators for dynamical systems [3, 4, 5]. In these settings, priors are often introduced to stabilize rollout, improve sample efficiency, or encode known structure suggested by measurement geometry or domain knowledge. The contribution here is complementary rather than competing. The paper does not propose a universal scientific world model; it proposes a standardized, low-cost preflight workflow for deciding whether a candidate prior should be carried into larger runs under a specified model condition.

A third strand includes causal and structural representation learning [6]. That literature seeks latent variables or relational structure aligned with underlying mechanisms, and it often asks whether learned representations capture more than predictive convenience. The present manuscript adopts some of the same caution about interpretation, but its goal is narrower. The workflow compares predictive priors under matched controls and asks whether observed gains support topology-specific attribution. It is not a causal discovery method, and it does not attempt to identify a true interaction graph from data alone.

A fourth strand concerns model-selection, ablation, and diagnostic workflows in machine learning and computational science. These approaches emphasize controlled baselines, matched comparisons, calibration, ablation, and concise reporting to guide computational decisions. The present protocol belongs most directly in this tradition. Its emphasis is on matched prior comparisons, persistence checks, and optional audits that produce an actionable prior recommendation before larger sweeps are undertaken, rather than on claiming state-of-the-art benchmark performance.

Taken together, these literatures motivate the need for the present study while also clarifying its boundary. This paper should be read as a prior-selection and attribution workflow for node-wise scientific latent dynamics: it draws on graph-based inductive bias, latent dynamics modeling, and diagnostic practice, but it does not claim a new graph architecture, a causal discovery result, or a state-of-the-art benchmark.

3. Problem formulation

We consider node-wise physical or scientific trajectory data $X_{1:T}$, where each frame $X_t \in \mathbb{R}^{N\times d}$ contains d observed features for each of N nodes. A dataset adapter provides these trajectories, a candidate graph or graph Laplacian $L \in \mathbb{R}^{N\times N}$, and metadata describing the graph source, the underlying simulator or dataset, and whether the graph is known, approximate, constructed, or data-derived. The predictive model maps observations into learned latent node states $H_t \in \mathbb{R}^{N\times m}$ and trains a transition model to forecast future dynamics.

The relevant object for interpretation is the model condition. In this manuscript, the model condition comprises the encoder, transition model, latent dimension, prior placement, optimizer settings, prior strength, training budget, and rollout evaluation horizon. A preflight result is therefore not a dataset-level truth statement. It is a recommendation tied to the tested model condition and must be reconsidered if the architecture, prior location, regularization strength, budget, or evaluation horizon changes.

The central question is the following:

> Under this model condition and training budget, does a candidate prior improve rollout prediction, and is the improvement better attributed to candidate topology, generic graph-frequency smoothing, calibrated temporal smoothing, or a low-budget regularization effect?

This formulation is intentionally narrower than causal discovery, universal physics modeling, or benchmark optimization. The objective is to reduce unnecessary full-training compute, avoid false topology-specific attribution, and decide whether the next step should be no prior, a candidate graph prior, calibrated temporal smoothing, or audit mode. Throughout the reported experiments, rollout quality is summarized at fixed horizons, especially $H = 16$ and $H = 32$, with $H = 32$ serving as the principal manuscript readout.

Two non-goals follow directly from this framing. First, the protocol does not claim that a useful candidate graph is physically true. Second, it does not claim universal coverage across scientific systems. The experiments are therefore presented as representative node-wise physical/scientific dynamics regimes rather than as a universal physics prediction study.

4. Mathematical and computational framing

The protocol separates prior utility from topology-specific attribution by evaluating several prior families under the same model condition, with the graph-spectral viewpoint following standard references in spectral graph theory [2]. The baseline condition is `none`, in which the latent dynamics model is trained without an auxiliary prior. The principal candidate graph prior is graph Laplacian smoothness on learned node-wise latent states. For a latent matrix H, the graph penalty is

$$R_G(H; L) = \mathrm{Tr}(H^\top L H) \tag{1}$$

If $L = U\Lambda U^\top$ is the eigendecomposition of the Laplacian and $H = UA$, then

$$\mathrm{Tr}(H^\top L H) = \mathrm{Tr}(A^\top \Lambda A) = \sum_k \lambda_k \|a_k\|_2^2 \tag{2}$$

This identity clarifies why graph-prior gains require caution. The candidate graph prior penalizes high graph frequencies more strongly, so improved rollout may reflect useful scientific structure, but it may also reflect generic smoothing in the graph spectral domain.

To probe that distinction, the key attribution baseline is the spectrum-matched permuted graph control. With permutation matrix P, the control Laplacian is

$$L_{\mathrm{perm}} = P^\top L P \tag{3}$$

This transformation preserves the eigenvalues of L, and hence the scale of graph-frequency smoothing, while disrupting node-label semantics. Within this protocol, if the candidate graph prior improves over the no-prior baseline but not over the spectrum-matched permuted graph control, the evidence supports predictive smoothing rather than topology-specific attribution. Random-graph controls can be informative secondary checks, but they are weaker because they do not preserve the full Laplacian spectrum.

The protocol also includes a graph-free baseline,

$$R_T(H_{1:T}) = \sum_t \|H_{t+1} - H_t\|_F^2 \tag{4}$$

which we refer to as calibrated temporal smoothing when its effective regularization scale has been matched to the candidate graph prior. Calibration is necessary because raw temporal losses can be much smaller than graph-Laplacian losses. If α_G denotes the graph-prior weight and R_G^0 and R_T^0 denote initial auxiliary-loss values measured under the same calibration pass, the temporal weight is chosen as

$$\alpha_T = \frac{\alpha_G R_G^0}{R_T^0 + \varepsilon} \tag{5}$$

so that

$$\alpha_T R_T^0 \approx \alpha_G R_G^0 \tag{6}$$

This calibration does not assert that graph and temporal priors are equivalent. It ensures that the comparison asks a fairer question: at matched initial regularization strength, does the candidate graph add value beyond graph-free temporal smoothing? The same framing also clarifies why, in our evidence, graph versus none alone is insufficient for topology-specific attribution. That stronger interpretation requires explicit comparison with the spectrum-matched permuted graph control and with calibrated temporal smoothing.

5. Model-conditioned preflight protocol

The model-conditioned preflight protocol is designed as a staged computational screen rather than a substitute for full training. It begins with diagnostics that summarize the candidate graph and the observed dynamics, proceeds to controlled mini-training runs, and escalates to longer budgets or latent audits only when the quick result warrants additional analysis.

Operationally, the workflow has four stages. First, trajectories, candidate graph metadata, and the corresponding Laplacian are loaded from a dataset adapter. Second, the prior family set is defined: `none`, the candidate graph prior, the spectrum-matched permuted graph control, optional random-graph controls, and calibrated temporal smoothing when a graph-versus-temporal comparison is needed. Third, models are trained under a screening budget and, when justified, under a longer budget; rollout is evaluated at fixed horizons. Fourth, if checkpointed latent traces are available, audit mode examines whether rollout gains coincide with smoother or more low-frequency latent temporal deltas in the candidate graph basis.

Figure 1 summarizes this workflow.

Audit mode is used when checkpointed latent traces are available; it asks whether predictive gains coincide with latent behavior aligned with the candidate graph and provides the strongest available evidence for topology-aligned support. Figure 2 shows how these comparisons lead to topology-specific attribution only when the relevant controls have been cleared.

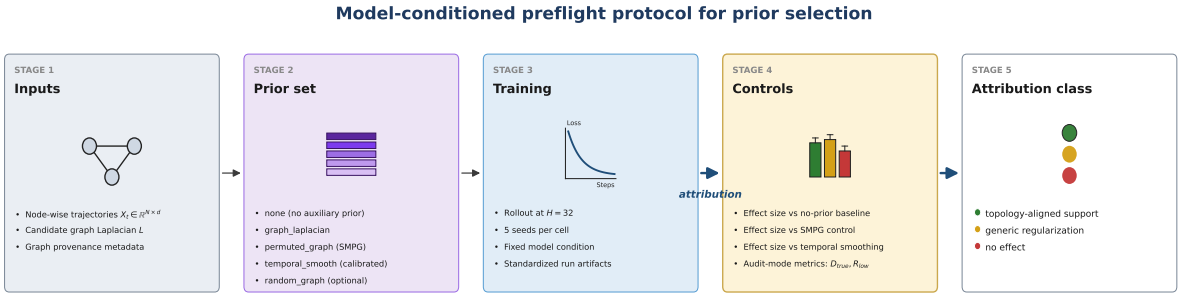

Figure 1: Model-conditioned preflight workflow. The protocol converts prior selection into a staged computational decision. Node-wise trajectories and a candidate graph first pass through raw diagnostics, then matched-prior comparisons against the spectrum-matched permuted graph (SMPG) control and calibrated temporal smoothing. An optional audit mode checks whether predictive gains coincide with smoother and more low-frequency learned latent temporal deltas in the candidate-graph basis. The output is an actionable prior recommendation under the tested model condition rather than a claim that the candidate graph is physically true.

The resulting labels are listed in Table 1, and the prior families on which they depend are summarized in Table 2. Taken together, these tables define the vocabulary of the protocol and the corresponding actionable prior recommendation associated with each outcome.

Table 1: Protocol labels and corresponding actionable prior recommendations

display label	evidence pattern	recommended action	claim allowed	claim boundary
no prior gain	No tested prior improves meaningfully over the no-prior model.	Skip the prior for larger runs unless there is a separate scientific reason to test it.	The tested prior was not useful under the model condition.	Does not imply the prior is useless for all models or budgets.
generic smoothing	The candidate graph improves over no-prior training but does not beat the spectrum-matched permuted graph control or comparable smoothing controls.	Treat the result as smoothing-like and test cheaper or simpler controls first.	Graph-style regularization helped prediction.	Do not claim topology-specific attribution.
temporal sufficient	Calibrated temporal smoothing matches or beats the candidate graph prior.	Prefer calibrated temporal smoothing unless topology-specific attribution is itself the objective.	Graph-free temporal smoothing explains the observed gain.	Do not claim that graph structure is necessary.
candidate topology	The candidate graph beats no-prior training, the spectrum-matched permuted graph control, and calibrated temporal smoothing under the tested budget.	Carry the candidate graph forward with more seeds, stronger controls, and audit where possible.	The result supports topology-specific attribution under the tested condition.	It does not prove that the candidate graph is the true physical interaction graph.
topology-aligned audit	Rollout gains coincide with smoother or more low-frequency learned latent temporal deltas in the candidate graph basis.	Use as the strongest preflight support for continuing the candidate graph.	Audit evidence supports topology-aligned latent smoothing under the tested condition.	The result remains model-conditioned, weight-conditioned, and budget-conditioned.

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display label	evidence pattern	recommended action	claim allowed	claim boundary
overconstrained	A prior harms rollout relative to no-prior training and controls.	Reduce strength, revise prior placement, or skip the prior.	The tested prior suppressed useful predictive variation.	This does not rule out weaker or differently placed priors.
inconclusive	Required controls, seeds, or audit artifacts are insufficient.	Run the missing controls or avoid stronger claims.	Evidence is insufficient for a stronger label.	Do not over-interpret incomplete comparisons.

Table 2: Prior families and attribution roles

prior family	definition	attribution role	manuscript use
`none`	No auxiliary prior.	Establishes whether any prior improves rollout.	Baseline for all preflight recommendations.
`graph_laplacian`	Candidate-graph Laplacian smoothness on learned node-wise latents, $\mathrm{Tr}\left(H^\top L H\right)$.	Tests whether candidate graph-frequency smoothing helps the tested model.	Main prior under evaluation, not introduced as a new prior.
`permuted_graph`	Spectrum-matched permuted Laplacian, $L_{\mathrm{perm}} = P^\top L P$.	Preserves Laplacian spectrum while removing node-label semantics.	Primary control for topology attribution in this protocol.
`random_graph`	Optional graph with matched coarse size or edge statistics.	Tests whether an unrelated graph can produce similar smoothing.	Secondary control; weaker than the spectrum-matched permuted graph control.
`temporal_smooth`	Graph-free smoothness on learned latent temporal deltas, e.g. $\sum_t \lVert H_{t+1} - H_t \rVert_F^2$.	Tests whether temporal regularization explains candidate graph gains.	Must be calibrated when compared with graph priors.
future priors	Planned auxiliary baselines for additional scientific structure.	Broadens the preflight family set.	Mentioned as future extensions, not current core evidence.

Implementation Overview

In practice, the preflight workflow is executed through an adapter-based preflight script. Each adapter is responsible for dataset-specific trajectory loading and for supplying the candidate graph information needed by the model-conditioned comparison. Depending on the regime, this means either loading a candidate graph Laplacian directly or constructing the candidate graph Laplacian from dataset-specific metadata before the matched prior comparisons are run.

This design keeps the comparison logic uniform across datasets. Once trajectories, the candidate graph Laplacian, and the associated metadata have been exposed through the adapter interface, the same preflight routine can evaluate no-prior training, the candidate graph prior, the spectrum-matched permuted graph control, optional random-graph controls, and calibrated temporal smoothing under a shared model condition. The resulting artifacts are written in a standardized form: `run_config.json` records the run specification, `summary.csv` provides the compact metric summary, and `preflight_report.md` provides the human-readable interpretation and actionable prior recommendation.

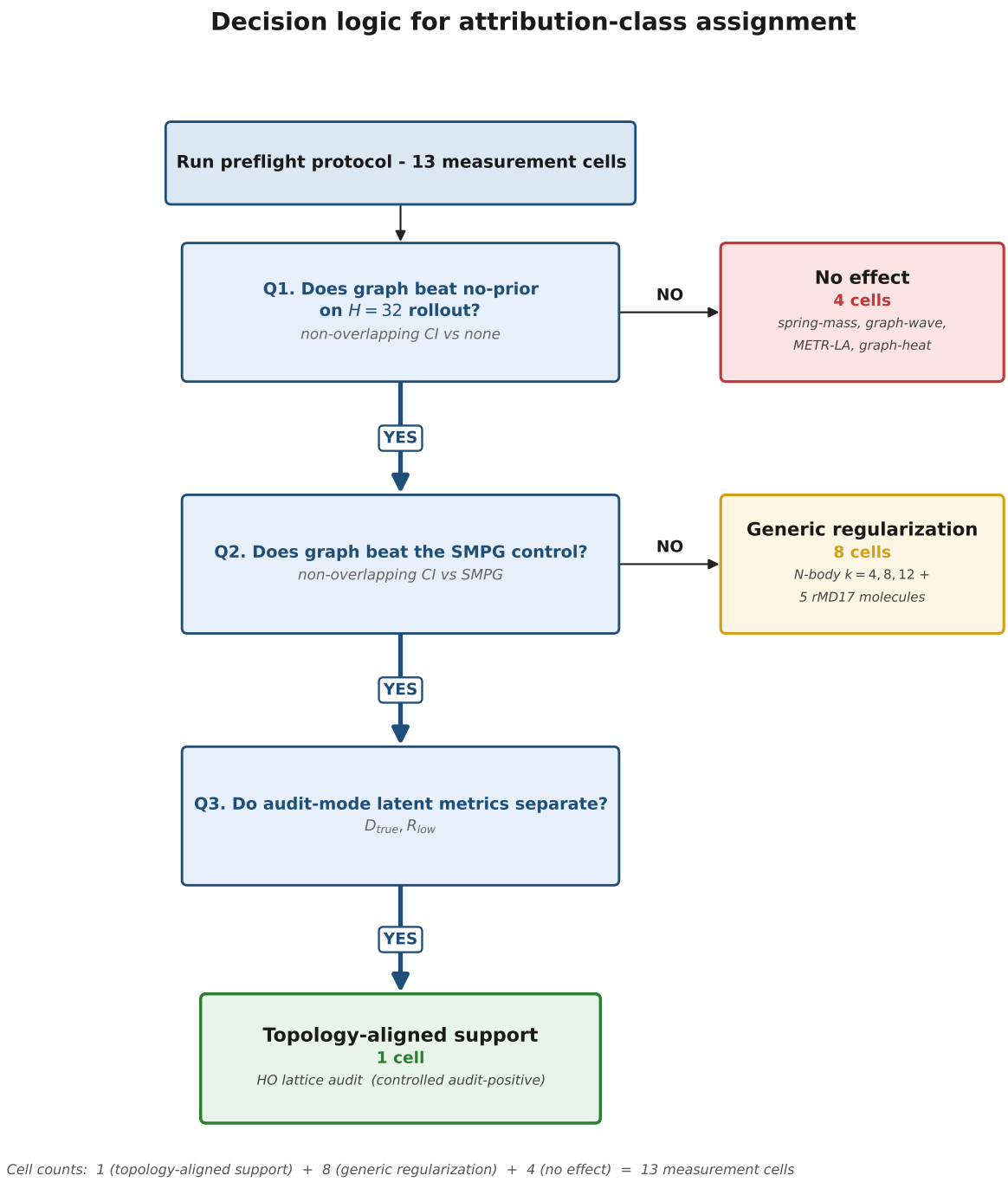

Figure 2: Decision logic for attribution-class assignment. The decision tree separates predictive utility from topology-specific attribution. A candidate graph must first improve over no-prior training, then beat the spectrum-matched permuted graph (SMPG) control under the operational non-overlapping confidence-interval criterion before it supports topology-specific attribution under the tested condition. Audit-mode latent metrics, when available, provide the strongest evidence of topology-aligned support. Cells that improve over no-prior but do not separate from SMPG are classified as generic regularization; cells that fail to improve over no-prior are classified as no-effect.

The implementation is also intended to make extension straightforward without changing the underlying methodology. New dataset support is added through adapters rather than by rewriting the core preflight procedure, so the same low-cost screening can be applied before larger sweeps, longer budgets, or expanded seed studies. In that sense, the implementation overview is part of the methodological contribution: it operationalizes standardized low-cost screening while keeping dataset-specific loading and candidate graph construction isolated from the matched prior comparisons themselves.

6. Experimental regimes and implementation

The experiments are presented as a stress test of the preflight methodology rather than as a leaderboard. All reported values are drawn from existing completed runs and summaries documented in the accompanying source materials; no new experiments are introduced in this revision. The purpose of the experimental section is to ask whether a single model-conditioned protocol can return sensible decisions across representative node-wise physical/scientific dynamics regimes.

The benchmark set spans controlled oscillator lattices, spring-mass systems, graph-wave and graph heat lattices, a graph low-frequency stress test, approximate n-body interactions, molecular dynamics summaries related to MD17/rMD17 [7, 8], and a traffic forecasting case connected to METR-LA as popularized by DCRNN [9]. These regimes differ not only in the underlying dynamics but also in candidate graph provenance: some graphs are simulator-defined, some are approximate physical constructions, some are synthetic stress tests, and some are data-derived. This heterogeneity is intentional because the provenance of the candidate graph is part of the interpretation boundary.

Implementation settings are fixed within each model condition. The screening budget used here is typically five epochs; longer-budget runs use twenty epochs where available. Audit mode is used only when checkpointed latent traces are available. Rollout comparisons focus on $H = 32$, which is the central summary statistic used throughout the figures and tables. Table 3 summarizes the regimes and the role each plays in the manuscript, while detailed file-level traceability is retained in Appendix D.

Table 3: Benchmark regimes used for preflight stress testing

regime	graph source	role in manuscript	key preflight readout
HO lattice	true lattice	Controlled topology-specific audit-positive case.	The candidate graph improves H32 rollout and the Cycle 8 audit shows smoother, more low-frequency latent deltas on the true graph.
Spring-mass lattice	true spring lattice	Second-order mechanics and temporal-baseline check.	The quick candidate graph gain is explained by calibrated temporal smoothing; the standard budget favors no-prior training.

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regime	graph source	role in manuscript	key preflight readout
Graph-wave lattice	true wave lattice	PDE-like propagation and temporal-baseline check.	The quick candidate graph gain is smoothness-like; calibrated temporal smoothing slightly beats the candidate graph; the standard budget favors no-prior training.
N-body distance	initial distance-kNN weighted graph	Approximate long-range interaction graph.	The quick candidate graph beats no-prior training, the spectrum-matched permuted graph control, and calibrated temporal smoothing; the standard budget favors the spectrum-matched permuted graph control.
METR-LA correlation graph	data-derived `correlation_topk` sensor graph	Real traffic application case.	Raw alignment does not translate into rollout gain; the candidate graph is worse than no-prior training and the spectrum-matched permuted graph control.
Graph heat lattice	true generator graph	Cautionary graph-generated case.	The true generator graph does not help the tested latent model; the candidate graph harms H32 rollout.
Graph low-frequency lattice	constructed low-frequency Laplacian basis	Synthetic alignment stress test.	Strong raw alignment does not guarantee quick topology-specific attribution; budget dependence appears in follow-up analysis.
rMD17 molecular dynamics summaries	molecular cutoff graphs	Earlier graph-style smoothing and specificity stress test.	Graph-style priors improve rollout, but exact molecular topology-specific attribution is weak against controls.

7. Results

We organize results around the three operational attribution classes introduced in Section 5. Across 13 measurement cells spanning seven regimes, the protocol identifies one controlled audit-positive / topology-aligned support case, eight generic-regularization cells, and four no-effect cells. Table 4 summarizes; the rest of this section walks through each class with detailed numbers from Table 5.

7.1. Controlled audit-positive / topology-aligned support: HO lattice audit

The HO lattice audit is the only controlled audit-positive / topology-aligned support cell in our evidence. Its rollout point estimates favor the candidate graph (H32 $= 0.0169 \pm 0.0054$ versus SMPG 0.0198 ± 0.0059 and random 0.0199 ± 0.0062), but the 95% rollout confidence intervals overlap across the three arms (graph $[0.0102, 0.0236]$; SMPG $[0.0125, 0.0271]$; random $[0.0122, 0.0275]$). We therefore do not treat rollout point-estimate ordering alone as evidence of topology-aligned support. The stronger evidence comes from audit-mode latent metrics, which are separated in the expected direction: the candidate-graph latent temporal delta norm $D_{\text{true}}(\Delta H)$ is 2.81 ± 0.04 versus 3.51 ± 0.11 for SMPG, and the low-frequency fraction R_{low} at K=4 is 0.187 ± 0.012 versus 0.122 ± 0.009 for SMPG. The agreement

Table 4: Headline attribution table across 13 measurement cells. Classification key: `controlled_audit_positive` denotes the controlled audit-positive / topology-aligned support class (HO audit only); `generic_regularization` denotes cells where graph improves over no-prior but does not separate from SMPG control under non-overlapping CI criterion; `no_effect` denotes cells with no meaningful candidate-graph effect.

regime	cell	n_seeds	classification	effect vs none	effect vs permuted
ho_audit_5seed	ho_lattice_audit	5	controlled_audit_positive	N/A	3.083
nbody_robustness_5seed	distance_k_04/standard_ep20	5	generic_regularization	6.014	0.535
nbody_robustness_5seed	distance_k_08/standard_ep20	5	generic_regularization	4.768	−0.370
nbody_robustness_5seed	distance_k_12/standard_ep20	5	generic_regularization	4.824	0.906
rmd17_cycle2_3seed	aspirin	3	generic_regularization	8.367	0.147
rmd17_cycle2_3seed	ethanol	3	generic_regularization	5.196	−2.935
rmd17_cycle2_3seed	malonaldehyde	3	generic_regularization	3.979	−0.445
rmd17_cycle2_3seed	naphthalene	3	generic_regularization	6.785	0.191
rmd17_cycle2_3seed	toluene	3	generic_regularization	2.359	−0.289
graph_heat_lattice	legacy_ep5_no_quick_tag	1	no_effect	N/A	N/A
graph_wave_5seed	standard_ep20	5	no_effect	0.352	−0.469
metr_la_3seed	corr_T2000_train160_ep5	3	no_effect	−0.736	−0.132
spring_mass_5seed	standard_ep20	5	no_effect	0.088	0.189

between rollout point-estimate ordering and latent audit-metric separation forms our operational basis for classifying HO as audit-positive. We bound this interpretation: it is a controlled audit-positive case in a setting where the candidate graph corresponds exactly to the physical Laplacian; it is not a topology-discovery result and is not evidence that distance-kNN or other approximate graphs in our other regimes would behave similarly.

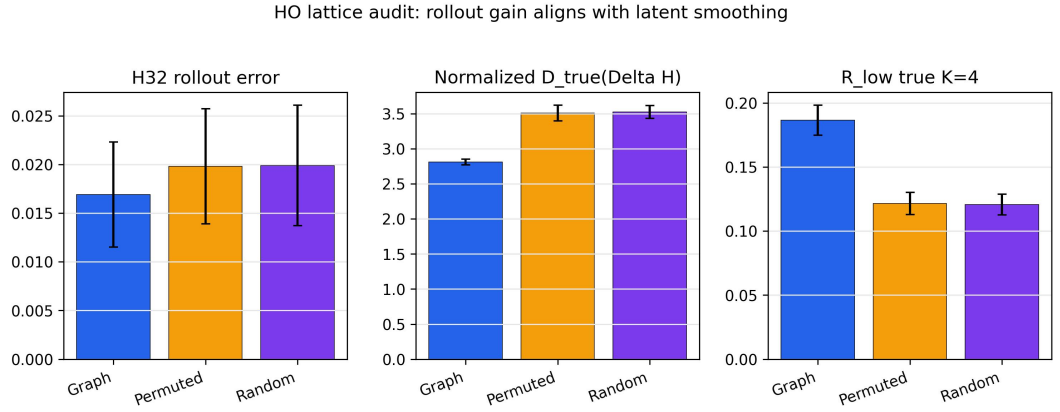

Figure 3: HO lattice audit: controlled audit-positive case. The candidate-graph latent temporal deltas are smoother and more low-frequency in the candidate-graph basis than the SMPG and random-graph controls. Rollout confidence intervals overlap across the three arms; the audit-positive classification rests on the agreement between rollout point-estimate ordering and latent audit-metric separation.

7.2. Generic regularization across synthetic and molecular dynamics

Eight cells form the generic-regularization block: three N-body standard cells at distance-k connectivities $k = 4$, 8, 12, and five rMD17 Cycle-2 molecules.

N-body distance-k standard cells.. Figure 4 shows the standard-budget rollout for the three N-body cells. In all three, the candidate graph substantially improves rollout error over no-prior training. At H32, the candidate graph achieves 0.064 ± 0.032 (k=4), 0.081 ± 0.033

(k=8), and 0.077 ± 0.031 (k=12), versus no-prior baselines of 0.151 ± 0.013, 0.152 ± 0.012, and 0.144 ± 0.012, respectively. Effect sizes against the no-prior baseline are 6.01, 4.77, 4.82 with non-overlapping CIs in all three cases. Against the SMPG control, however, effect sizes fall to 0.54, −0.37, 0.91 with confidence-interval overlap in every cell, and against calibrated temporal smoothing they fall to 0.14, −0.80, −1.27 with overlap. Under the operational CI criterion, the candidate graph is not statistically separable from the SMPG control in any of the three connectivities. We therefore attribute the graph-vs-none improvement to generic regularization rather than to identifiable distance-kNN topology.

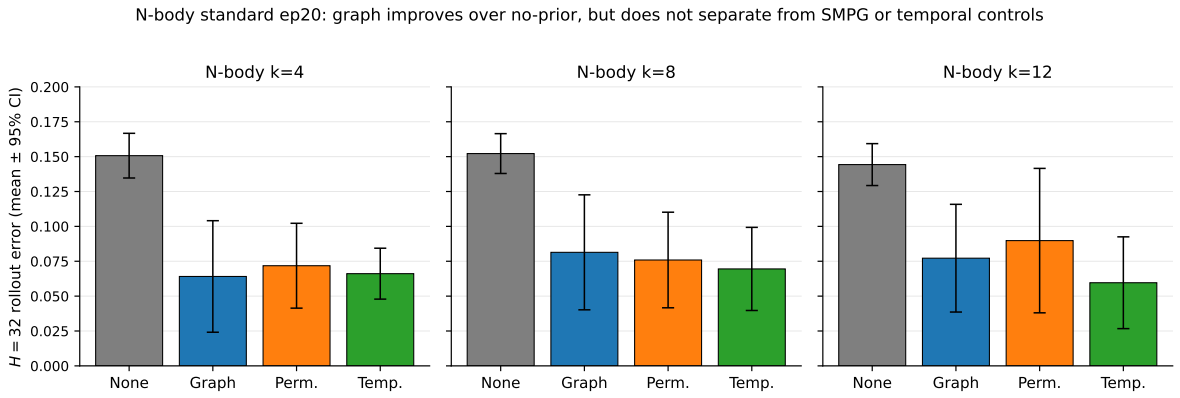

Figure 4: N-body standard-budget rollout (20 epochs, 5 seeds, H=32). In all three connectivities (k=4, 8, 12) the candidate distance-kNN graph (Graph) substantially reduces rollout error against the no-prior baseline (None), but the spectrum-matched permuted graph control (Perm.) and calibrated temporal smoothing (Temp.) achieve comparable or slightly better mean rollout. Error bars are 95% confidence intervals (t-distribution, df=4). Under the operational non-overlapping CI criterion, the candidate graph is not statistically separable from either control in any of the three cells.

rMD17 molecular-dynamics cells.. Five rMD17 Cycle-2 molecules provide independent real-chemistry evidence with the same pattern. In all five molecules, the node-wise graph prior reduces H32 over no-prior `gnn_node` training: effect sizes against the no-prior baseline range from 2.36 (toluene) to 8.37 (aspirin), spanning the same magnitude regime as N-body. However, in zero of the five molecules does the candidate graph achieve non-overlapping CIs against the SMPG control or against the random-graph control. In three molecules (ethanol, malonaldehyde, toluene), permuted or random controls match or slightly outperform the candidate molecular cutoff graph at H32 (e.g., ethanol: graph 0.0449 ± 0.0036 versus permuted 0.0388 ± 0.0081; effect size against SMPG = −2.94, with overlapping CIs). Because these rMD17 results are 3-seed Cycle-2 data without a calibrated temporal-smoothing arm, we use them as cross-domain corroborating evidence rather than as the sole basis for the attribution rule. They extend the generic-regularization classification from synthetic N-body data to real molecular dynamics across five distinct organic molecules.

7.3. No-effect cells

Four cells show no meaningful candidate-graph effect at standard budget. Spring-mass lattice has graph H32 = 0.0800 ± 0.0379 versus none = 0.0815 ± 0.0136 (effect 0.09, CI overlap). Graph-wave lattice has graph 0.0782 ± 0.0324 versus none 0.0833 ± 0.0249 (effect

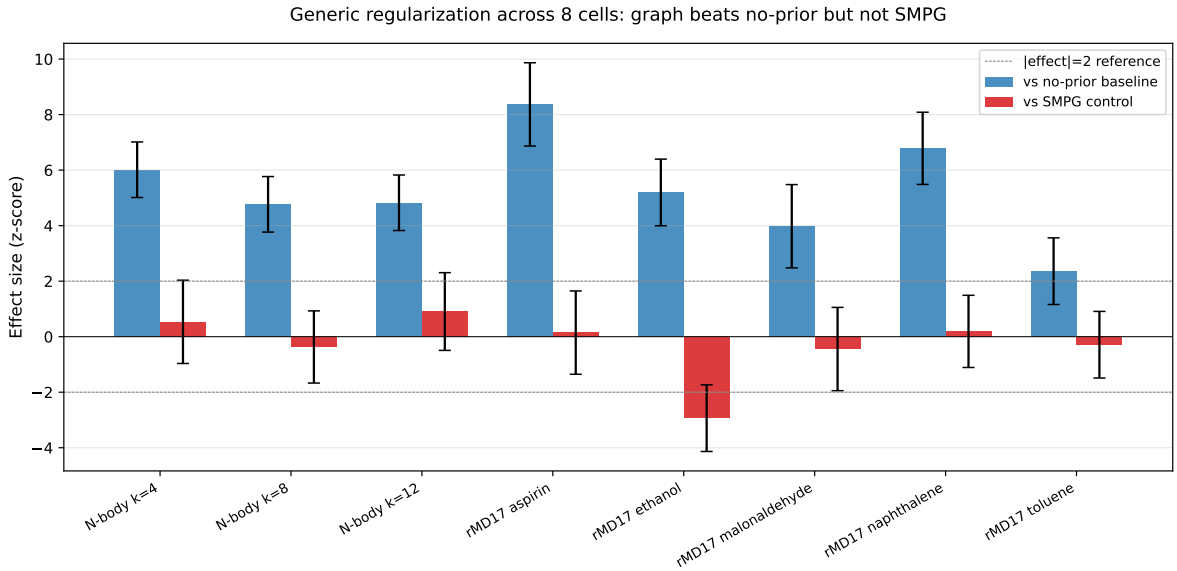

Figure 5: Generic-regularization cells: effect sizes against no-prior baseline (left bars) versus against SMPG control (right bars), with 95% confidence intervals. In all eight cells the candidate graph substantially improves rollout over no-prior training (left bars), but the improvement does not separate from the SMPG control (right bars) under the operational non-overlapping CI criterion.

0.35, overlap). METR-LA correlation graph has graph 0.5243 ± 0.2845 versus none 0.4034 ± 0.0609 (effect -0.74; the candidate graph performs worse than no-prior on average, although CIs overlap). Graph-heat lattice is retained as a legacy single-seed cautionary cell in the no-effect block, although it is not part of the multiseed subset: graph H32 = 0.6902 versus none = 0.6036, so the candidate graph performs worse under the tested model condition. The protocol's output for these cells is "do not carry the candidate graph forward in this model condition", which we treat as the appropriate honest recommendation.

7.4. Methodological cross-validation

As representative methodological cross-checks, we run node-order and transition-design diagnostics in selected regimes.

Node-order sanity check.. In spring-mass and N-body k=4, we run a node-order sanity check that probes whether the model's prediction pathway is sensitive to graph–data alignment. Three conditions are compared: (A) baseline (data and graph both unpermuted), (B) data and graph permuted consistently with the same permutation (permutation-equivariance check), and (C) data permuted while the candidate graph is left unchanged (alignment-break stress). In spring-mass, the H32 values are A = 0.62 ± 0.17, B = 0.62 ± 0.17, C = 0.74 ± 0.35. The B = A equality confirms permutation equivariance; the C value is only $\sim 19\%$ worse than A despite breaking graph–data alignment, indicating that the prediction pathway weakly utilizes the candidate graph in this model condition. A WARNING flag is raised. The same WARNING is raised on N-body k=4. This independent diagnostic is consistent with the generic-regularization classification: the model condition does not strongly exploit topology, regardless of prior strength.

Transition-architecture redesign.. As a robustness probe, we also tested whether the generic-regularization pattern was an artifact of the pooled-latent transition design. Replacing the transition with a graph-aware SAGEConv-plus-residual-MLP module stabilized by LayerNorm did not rescue topology attribution: spring-mass remained no-effect (effect against SMPG = 0.005), and N-body k=4 remained generic regularization (effect against no-prior = 3.79; effect against SMPG = 0.08, CI overlap). The node-order sanity WARNING also persisted. We therefore treat generic-regularization dominance as a model-condition finding rather than a trivial artifact of the original transition head.

7.5. Summary

Across 13 measurement cells, the protocol produces three operational attribution outputs: one controlled audit-positive / topology-aligned support case (HO audit), eight generic-regularization cells (three N-body standard cells plus five rMD17 molecules), and four no-effect cells. Without SMPG and calibrated temporal-smoothing controls, the eight generic-regularization cells would have appeared as graph-versus-no-prior successes; the protocol redirects this finding into the appropriate attribution class. We discuss the implications in Section 8.

Table 5: Unified effect-size table. Effect sizes are computed as (comparator_mean - graph_mean) / (std_graph / sqrt(n_graph)). Positive values indicate the candidate graph achieves lower H32 rollout error. CI overlap labels are based on 95% confidence intervals.

regime	cell	n_seeds	mean H32 graph	effect vs none	CI vs none	effect vs SMPG	CI vs SMPG	effect vs random	CI vs random	effect vs temporal	CI vs temporal	classification
`ho_audit_5seed`	`ho_lattice_audit`	5	0.0169	N/A	N/A	3.083	non-overlap	2.745	non-overlap	N/A	N/A	`controlled_audit_positive`
`nbody_robustness_5seed`	`distance_k_04/standard_ep20`	5	0.0641	6.014	non-overlap	0.535	overlap	N/A	N/A	0.139	overlap	`generic_regularization`
`nbody_robustness_5seed`	`distance_k_08/standard_ep20`	5	0.0814	4.768	non-overlap	−0.370	overlap	N/A	N/A	−0.801	overlap	`generic_regularization`
`nbody_robustness_5seed`	`distance_k_12/standard_ep20`	5	0.0772	4.824	non-overlap	0.906	overlap	N/A	N/A	−1.265	overlap	`generic_regularization`
`rmd17_cycle2_3seed`	aspirin	3	0.0306	8.367	non-overlap	0.147	overlap	−0.013	overlap	N/A	N/A	`generic_regularization`
`rmd17_cycle2_3seed`	ethanol	3	0.0449	5.196	overlap	−2.935	overlap	−2.358	overlap	N/A	N/A	`generic_regularization`
`rmd17_cycle2_3seed`	malonaldehyde	3	0.0346	3.979	overlap	−0.445	overlap	−0.480	overlap	N/A	N/A	`generic_regularization`
`rmd17_cycle2_3seed`	naphthalene	3	0.0210	6.785	non-overlap	0.191	overlap	0.381	overlap	N/A	N/A	`generic_regularization`
`rmd17_cycle2_3seed`	toluene	3	0.0323	2.359	overlap	−0.289	overlap	−0.247	overlap	N/A	N/A	`generic_regularization`
`graph_heat_lattice`	`legacy_ep5_no_quick_tag`	1	0.6902	N/A	non-overlap	N/A	non-overlap	N/A	N/A	N/A	N/A	`no_effect`
`graph_wave_5seed`	`standard_ep20`	5	0.0782	0.352	overlap	−0.469	overlap	N/A	N/A	−0.559	overlap	`no_effect`
`metr_la_3seed`	`corr_T2000_train160_ep5`	3	0.5243	−0.736	overlap	−0.132	overlap	N/A	N/A	−0.089	overlap	`no_effect`
`spring_mass_5seed`	`standard_ep20`	5	0.0800	0.088	overlap	0.189	overlap	N/A	N/A	−0.024	overlap	`no_effect`

Note. HO audit graph-vs-control effects use paired bootstrap deltas from the specificity audit because the audit cell has no no-prior arm. `controlled_audit_positive` denotes the controlled audit-positive class; HO audit is the only such case in our evidence.

8. Discussion

The main contribution of the manuscript is methodological. It provides a principled way to decide, under a fixed model condition, whether a graph-prior gain supports candidate topology, generic regularization, or no meaningful graph effect. In that sense, the protocol addresses a recurring problem in computational science: prior choice is consequential, but prior interpretation is easy to overstate.

The reported cases sharpen the distinction between predictive utility and topology-specific attribution. A candidate graph beating the no-prior baseline can be useful, but it has not yet shown that node-label structure matters. Across 13 measurement cells, the protocol assigns one cell to controlled audit-positive / topology-aligned support (HO lattice audit), eight cells to generic regularization (three N-body standard cells and five rMD17 molecules), and four cells to no-effect. The dominant pattern is generic regularization rather than topology-aligned support: in the eight generic-regularization cells, effect sizes against the no-prior baseline range from 2.4 to 8.4, but effect sizes against the SMPG control range

from -2.9 to 0.9 with confidence-interval overlap in every cell. Without SMPG controls, these eight cells would have been over-attributed as graph-versus-no-prior successes; the protocol redirects them into the appropriate attribution class.

Several model-condition observations emerge from this evidence. First, the prediction pathway in our tested architecture weakly exploits topology even when the candidate graph is structurally plausible. The node-order sanity check provides an independent diagnostic of this insensitivity: in spring-mass, the alignment-break stress is only $\sim 19\%$ worse than baseline, and N-body $k = 4$ produces the same warning-level diagnostic. Second, the generic-regularization classification is not a trivial artifact of the pooled-latent transition design: replacing the transition with a graph-aware SAGEConv-plus-residual-MLP module did not rescue topology attribution. We therefore treat generic-regularization dominance as a model-condition finding rather than a purely architectural failure.

The HO audit deserves special discussion. It is the only cell in our evidence where rollout point-estimate ordering and audit-mode latent metrics agree in the direction predicted by topology-aligned learning. Importantly, even here the rollout confidence intervals overlap; the audit-positive classification rests on the agreement between rollout ordering and latent-space audit metrics rather than on rollout CI separation. This is a deliberately conservative operational standard. The HO setting is also the only cell where the candidate graph corresponds exactly to the physical Laplacian, making it a controlled best case rather than a topology-discovery result.

The negative cases reinforce the same operational lesson. METR-LA diagnoses candidate-graph construction risk: the tested correlation-derived graph fails without ruling out official road adjacency, road-distance graphs, or learned traffic graphs. Graph heat diagnoses prior-form risk: even a true generator graph can fail when static Laplacian smoothing is mismatched to diffusion-like transition dynamics. Spring-mass and graph-wave show how the same workflow can recommend against carrying a candidate graph forward when it fails to clear the baseline. The value of the protocol, therefore, lies as much in ruling out unsupported interpretations as in identifying promising priors.

The practical value is attribution control: the protocol turns graph-prior successes, failures, and heterogeneous outcomes into diagnostic categories with explicit claim boundaries. Table 6 summarizes the principal claims, their evidentiary basis, and the boundaries that accompany them. Detailed case-level traceability is retained in Appendix Appendix D.

Table 6: Claims, evidence, and boundaries

manuscript claim	**supporting evidence**	**key numbers**	**boundary**
The preflight protocol gives actionable prior recommendations.	Preflight benchmark summary and individual preflight reports.	Operational attribution labels include no gain, temporal sufficient, candidate topology-specific, generic smoothing, and audit-supported positive cases.	Recommendations apply only under tested model and budget conditions.
In this protocol, the spectrum-matched permuted graph control is necessary for topology-specific attribution.	Master decision rules and graph specificity table.	HO lattice lambda=0.1 beats the spectrum-matched permuted graph control at H32; many rMD17 cases do not.	In our evidence, a candidate graph beating no-prior training alone would have been over-attributed without SMPG controls.

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manuscript claim	supporting evidence	key numbers	boundary
Calibrated temporal smoothing changes candidate graph attribution.	Calibrated spring-mass, graph-wave, and n-body preflight reports.	Spring-mass temporal H32 0.5689 vs graph 0.5740; graph-wave temporal H32 0.5663 vs graph 0.5712; n-body graph H32 0.5433 vs temporal 0.6453.	Calibration is required because raw prior losses differ by scale.
Candidate-graph gains can be low-budget-only.	Quick/standard comparisons for spring-mass, graph-wave, and n-body.	Spring-mass graph H32 quick 0.5740 vs standard 0.1048, while standard no-prior is 0.0809; graph-wave standard no-prior is best; n-body standard spectrum-matched permuted graph control is best.	This does not imply that graph priors never help longer runs.
N-body standard cells show generic regularization, not topology-specific attribution.	N-body standard calibrated preflight reports.	H32 ($k = 4$ standard, 5 seeds) none 0.151 ± 0.013, graph 0.064 ± 0.032, permuted 0.072 ± 0.025, temporal 0.066 ± 0.015. Effect vs none 6.01 (non-overlap); effect vs SMPG 0.54 (overlap); effect vs temporal 0.14 (overlap).	Pattern persists at $k = 8, 12$. SMPG and temporal controls explain the apparent gain.
Audit mode can support topology-aligned latent smoothing.	Cycle 8 checkpointed HO lattice audit.	H32 graph 0.0169, permuted control 0.0198, random 0.0199; graph $D_{\text{true,norm}}(\Delta H)$ 2.8106 vs permuted control 3.5109; graph R_{low} at $K = 4$ 0.1866 vs 0.1216.	This is a controlled audit-positive case, not universal latent-alignment evidence.
Raw graph-dynamics alignment is not sufficient.	Cycle 6 diagnostics, METR-LA, graph low-frequency lattice, graph heat.	METR raw true $D_{dX,\text{norm}}$ 0.2380 vs permuted 0.3044, but graph H32 is worse than no-prior training; the graph low-frequency case also fails to establish quick topology-specific attribution.	Raw diagnostics are supporting probes, not decision rules.
Graph-generated dynamics do not guarantee candidate graph utility.	Graph heat lattice report and failure audit.	H32 none 0.6036, graph 0.6902, permuted control 0.6058.	The prior acts on learned latents, not directly on the simulator equation.

9. Limitations

The protocol is intentionally conservative. Its recommendations are model-conditioned and budget-conditioned: they apply to the tested encoder, transition model, latent dimension, prior placement, optimizer settings, prior strength, training budget, and rollout horizon. A recommendation should therefore be treated as a local computational conclusion rather than as a property of the dataset alone.

Several methodological boundaries follow from this framing. Candidate graphs in the reported regimes range from simulator-defined lattices to approximate physical constructions and data-derived correlation graphs, and those categories should not be interpreted in the same way. The spectrum-matched permuted graph control is the strongest control used here, but it does not exhaust every possible graph confound. Calibrated temporal smoothing is required for fair graph-versus-temporal comparison. Raw-coordinate diagnostics such as $D_L($ $DeltaX)$ can be useful supporting probes, but because the prior acts on learned latents, they are not sufficient decision rules.

A further limitation is that audit mode depends on checkpointed latent traces or saved latent artifacts. Where those artifacts are missing, mechanism claims must remain limited. Finally, the benchmark spans representative node-wise physical/scientific dynamics regimes rather than the whole space of scientific prediction problems. The manuscript should therefore not be read as introducing a universal physics predictor, a causal discovery method, or

a state-of-the-art benchmark.

Table 5 consolidates these boundaries in the form used throughout the draft.

Table 7: Limitations and claim boundaries

limitation or boundary	**boundary statement**	**implication**
Model conditioning	Recommendations apply under the tested encoder, transition model, prior placement, prior strength, optimizer, training budget, and rollout horizon.	Changing the model condition requires a new preflight interpretation.
Budget conditioning	Quick-mode gains can disappear under standard or longer training.	Quick positives should be validated before stronger claims.
Candidate graph ambiguity	Candidate graphs may be true, approximate, constructed, or data-derived.	A useful candidate graph prior does not prove physical truth.
Control requirements	In this protocol, topology-specific attribution requires beating the spectrum-matched permuted graph control and calibrated temporal smoothing.	Graph versus none is insufficient for topology-specific attribution.
Temporal calibration	Raw temporal prior losses may be much smaller than graph-prior losses.	Temporal baselines should be calibrated before attribution.
Raw diagnostics	Raw-coordinate smoothness can disagree with learned latent prior utility.	Raw $D_L(\Delta X)$ should not be used as a sufficient decision rule.
Audit artifacts	Latent audits require saved traces or checkpoints.	Missing artifacts limit mechanism claims.
Benchmark scope	Regimes are representative stress tests, not universal physics coverage.	Avoid universal predictor or SOTA benchmark framing.
Causal claims	The workflow compares predictive priors and controls.	Do not frame the result as causal discovery.
Prior novelty	The graph Laplacian prior is a tested prior family, not the paper's algorithmic novelty.	The contribution is the preflight protocol and attribution workflow.

10. Conclusion

This manuscript presents a model-conditioned preflight protocol for prior selection in node-wise scientific latent dynamics. By combining no-prior baselines, a candidate graph prior, the spectrum-matched permuted graph control, calibrated temporal smoothing, and audit mode when latent artifacts are available, the protocol turns prior choice into a structured computational decision rather than an ad hoc modeling preference.

Across 13 measurement cells in seven regimes, the dominant pattern is generic regularization rather than topology-aligned support. Eight of the 13 cells, spanning N-body synthetic data and five rMD17 molecules, show that graph priors substantially improve prediction over no-prior training but do not separate from the spectrum-matched permuted graph control under the operational non-overlapping confidence-interval criterion. The HO lattice audit is the single controlled audit-positive / topology-aligned support case, where rollout ordering and audit-mode latent metrics agree in the direction predicted by topology-aligned

learning. Four no-effect cells (spring-mass, graph-wave, METR-LA, graph-heat) show that graph priors do not always provide regularization benefit even under matched conditions. In our evidence, graph-versus-no-prior improvements would have been over-attributed without SMPG controls.

The contribution is therefore a practical workflow for prior selection and interpretation rather than a new graph prior, a universal physics predictor, a causal discovery method, or a state-of-the-art benchmark. The protocol's primary value is attribution control: it converts ambiguous "graph-vs-none" gains into structured operational classes that can guide subsequent computational decisions under a fixed model condition.

Data and Code Availability

This manuscript reports previously completed runs only and introduces no new experiments in this version. The preflight tool is implemented in the project repository as an adapter-based script in which dataset-specific adapters provide trajectories and candidate graph information, while the shared routine writes `run_config.json`, `summary.csv`, and `preflight_report.md` as standardized run artifacts. Code and manuscript assets are available at https://github.com/2GyounnnG/causalworld. A versioned archival snapshot will be provided upon publication or repository release.

Appendix A. Protocol details and operating modes

The protocol is best understood as a sequence of increasingly demanding questions. The first question is whether any prior helps over no-prior training. The second is whether the gain survives the spectrum-matched permuted graph (SMPG) control and calibrated temporal smoothing. Audit mode, when checkpointed latent traces are available, asks whether predictive gains coincide with latent behavior consistent with the candidate graph. The strongest label therefore requires agreement across utility, control comparisons, and latent-space evidence.

The labels in Table 1 should be interpreted as report-level decisions rather than as permanent properties of a dataset. They are intentionally conservative: `no_prior_gain` and `overconstrained` recommend against continuing the prior; `temporal_smoothing_sufficient` recommends a cheaper graph-free alternative; `candidate_topology_specific` supports topology-specific attribution under the tested condition; and `topology_aligned_latent_smoothing` requires audit evidence in addition to predictive gains.

Appendix B. Benchmark regimes and dataset-adapter notes

The benchmark set mixes graphs with different provenance. HO, spring-mass, graph-wave, and graph heat use true or simulator-defined lattices. The graph low-frequency regime uses a constructed spectral stress test. N-body uses an approximate distance-kNN graph. The rMD17 summaries use molecular cutoff graphs. METR-LA uses a correlation-derived sensor graph. This variation is intentional because candidate graph provenance is part of the interpretation boundary addressed by the preflight protocol.

Appendix C. Prior families and calibration formulas

The graph Laplacian prior on learned node-wise latent states is

$$R_G(H; L) = \mathrm{Tr}(H^\top L H) \tag{C.1}$$

The spectrum-matched permuted graph control is

$$L_{\mathrm{perm}} = P^T L P, \tag{C.2}$$

where P is a permutation matrix. This preserves Laplacian eigenvalues while disrupting node-label semantics.

The graph-free temporal smoothness prior is

$$R_T(H_{1:T}) = \sum_t \|H_{t+1} - H_t\|_F^2 \tag{C.3}$$

For calibrated prior-strength matching, let α_G be the graph-prior weight, and let R_G^0 and R_T^0 denote initial graph and temporal prior losses under the same calibration pass. The calibrated temporal weight is

$$\alpha_T = \frac{\alpha_G R_G^0}{R_T^0 + \varepsilon} \tag{C.4}$$

so that

$$\alpha_T R_T^0 \approx \alpha_G R_G^0 \tag{C.5}$$

Appendix D. Claim traceability and supporting evidence

This appendix provides a compact traceability table linking each major manuscript claim to the supporting experiment summaries, key numerical evidence, and interpretation caveats.

Table D.8: Claim traceability and supporting evidence

claim	**supporting evidence**	**key numbers**	**caveats**	**appears in**
The paper contributes a computational preflight protocol for model-conditioned prior selection.	`paper/jcs_outline.md`; `paper/jcs_draft_intro.md`; `paper/jcs_draft_methods.md`	Model condition includes encoder, transition model, prior placement, strength, training budget, and rollout horizon.	Not a new graph prior or SOTA benchmark.	Tables 1-2
In this protocol, the spectrum-matched permuted graph control is the key topology-specific attribution control.	`paper/jcs_draft_methods.md`; `paper/tables/evidence_map.md`	$L_{\mathrm{perm}} = P^\top L P$; HO lattice lambda=0.1 permuted-control delta 0.002866 at H32 with CI $[0.001070, 0.004210]$.	Random graph controls are secondary and not fully spectrum-matched.	Tables 2 and 4
Calibrated temporal smoothing can explain some candidate graph gains.	`paper/tables/evidence_map.md`; `paper/tables/table4_claims_evidence.md`	Spring-mass temporal H32 0.5689 vs graph 0.5740; graph-wave temporal H32 0.5663 vs graph 0.5712.	Requires calibrated prior strength; uncalibrated temporal comparisons are not enough.	Tables 1 and 4

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claim	supporting evidence	key numbers	caveats	appears in
N-body standard cells show generic regularization, not topology-specific attribution.	`paper/tables/evidence_map.md`; `paper/tables/table4_claims_evidence.md`	H32 ($k = 4$ standard, 5 seeds) none 0.151 ± 0.013, graph 0.064 ± 0.032, permuted 0.072 ± 0.025, temporal 0.066 ± 0.015. Effect vs none 6.01 (non-overlap); effect vs SMPG 0.54 (overlap); effect vs temporal 0.14 (overlap).	Pattern persists at $k = 8, 12$. SMPG and temporal controls explain the apparent gain.	Section 7; Tables 3-4
HO lattice/Cycle 8 is a controlled audit-positive / topology-aligned support case.	`paper/tables/evidence_map.md`; `paper/figure_data/fig6_latent_audit_positive_case.csv`	H32 graph 0.0169, permuted 0.0198, random 0.0199; $D_{\text{true,norm}}(\Delta H)$ graph 2.8106, permuted 3.5109, random 3.5237; R_{low} true $K = 4$ graph 0.1866, permuted 0.1216.	Rollout CIs overlap across arms; classification rests on agreement between rollout point-estimate ordering and audit metrics. Controlled best case; not a topology-discovery result.	Figure 3; Tables 3-4
METR-LA correlation graph demonstrates no graph gain despite raw alignment.	`paper/tables/evidence_map.md`; `paper/tables/table4_claims_evidence.md`	Raw true $D_{dX,\text{norm}}$ 0.2380, permuted 0.3044, random 0.4737; H32 none 0.4068, graph 0.4461, permuted 0.4337.	Uses `correlation_topk` graph, not official road adjacency.	Tables 3-5
Graph heat lattice shows true generator graph does not guarantee prior utility.	`paper/tables/evidence_map.md`; `paper/tables/table4_claims_evidence.md`	H32 none 0.6036, graph 0.6902, permuted 0.6058.	Applies to the tested latent model and prior setting.	Tables 3-5
rMD17 5-molecule Cycle-2 evidence corroborates the generic-regularization classification on real molecular dynamics.	`paper/tables/evidence_map.md`; `paper/tables/table4_claims_evidence.md`	5/5 molecules: graph reduces H32 vs `gnn_node` none (effect sizes 2.36-8.37). 0/5 molecules show non-overlapping CI vs SMPG control. 3/5 molecules: SMPG or random matches/outperforms graph.	3-seed Cycle-2 historical data; no calibrated temporal-smoothing arm, so used as corroborating cross-domain evidence.	Tables 3-4 and discussion

Appendix E. Reproducibility note and software interface

The present draft summarizes previously completed runs only. The software interface follows an adapter-based design in which dataset-specific trajectory loading and candidate graph Laplacian construction or loading are handled by adapters, while the core routine carries out the matched prior comparisons under a shared model condition. In this interface, `run_config.json` records the run specification, `summary.csv` provides the compact metric summary, and `preflight_report.md` provides the human-readable report and actionable prior recommendation. Companion documentation targets for this tool include `paper/tool_readme_draft.md` and `docs/graph_prior_preflight_tool.md`, alongside the command-line example below.

```
conda run -n causalworld python scripts/graph_prior_preflight_check.py \
  --dataset spring_mass_lattice \
  --prior-weight 0.1 \
  --epochs 5 \
```

```
  --train-transitions 96 \
  --eval-transitions 32 \
  --horizons 16 32 \
  --include-temporal-prior \
  --calibrate-prior-strength \
  --out-dir analysis_out/preflight_runs/spring_mass_lattice_ep5_temporal_calibrated
```

This command is included for interface clarity only; no new experiment is introduced in this manuscript version.

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